

SYDNEY TECHNICAL HIGH SCHOOL



PRELIMINARY HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 3

SEPTEMBER 2013

Mathematics

General Instructions

- Working time - 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in questions 11 to 18
- Start each question on a new page

Total marks - 82

Section 1 - 10 marks

Attempt Questions 1 – 10.
Allow about 15 minutes for this section.

Section 2 - 72 marks

Attempt Questions 11 – 18.
Allow about 105 minutes for this section.

Name : _____

Teacher : _____

Section 1

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

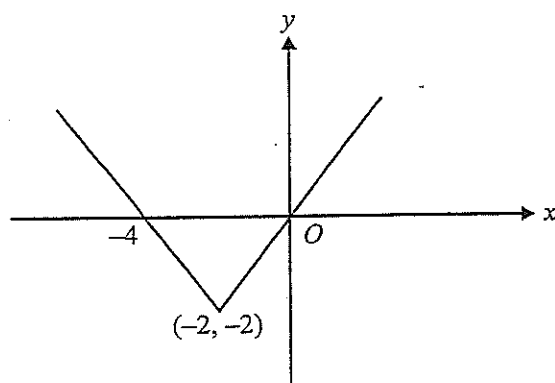
Use the multiple-choice answer sheet in your answer booklet for Questions 1 – 10.

Do not remove the multiple-choice answer sheet from your answer booklet.

1. If $x = -4$ is a root of the equation $2x^2 + kx + 4 = 0$, what is the value of k ?

- (A) 7
- (B) 8
- (C) 9
- (D) 10

2.



The rule of the function whose graph is shown is

- (A) $y = |x| - 4$
- (B) $y = |x - 2| + 2$
- (C) $y = |x + 2| - 2$
- (D) $y = |2 - x| - 2$

3. If $\sqrt{12} + \sqrt{3} = \sqrt{b}$ then

(A) $b = \sqrt{15}$

(B) $b = 3\sqrt{3}$

(C) $b = 15$

(D) $b = 27$

4. The x coordinates of the points of intersection of $y = x^2$ and $x + y = 6$ are the solutions of

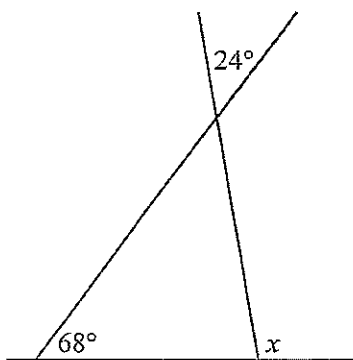
(A) $x^2 - x - 6 = 0$

(B) $x^2 + x - 6 = 0$

(C) $x^2 - x + 6 = 0$

(D) $x^2 + x + 6 = 0$

5.



The size of the angle x is

(A) 68°

(B) 88°

(C) 92°

(D) 112°

6. Given $y = a x^n$ then $\frac{dy}{dx} = ?$

(A) $a \times n \times x^{n-1}$

(B) $a \times n \times x^{n+1}$

(C) $n \times x^{n-1}$

(D) $a \times x^{n-1}$

7. Find the values of m for which $24 + 2m - m^2 \leq 0$

(A) $m \leq -4$ or $m \geq 6$

(B) $m \leq -6$ or $m \geq 4$

(C) $-4 \leq m \leq 6$

(D) $-6 \leq m \leq 4$

8. For $y = (4x + 1)(x + 2)^3$, $\frac{dy}{dx}$ is equal to

(A) $12(x + 2)^2$

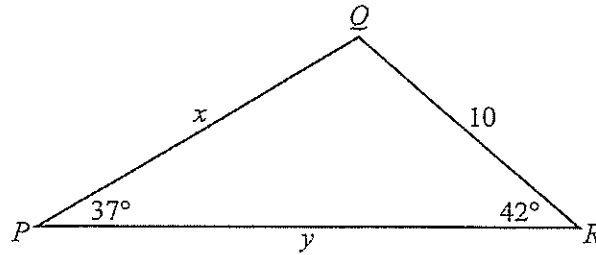
(B) $(x + 2)^2(16x + 11)$

(C) $3(x + 2)^2(4x + 1)$

(D) $(x + 2)^2(12x + 7)$

9. PQR is a triangle with side lengths x , 10 and y , as shown below.

In this triangle, angle RPQ = 37° and angle QRP = 42° .



Which one of the following expressions is correct for triangle PQR ?

- (A) $x = \frac{10}{\sin 37^\circ}$
- (B) $x = 10 \times \frac{\sin 42^\circ}{\sin 37^\circ}$
- (C) $y = 10 \times \frac{\sin 37^\circ}{\sin 101^\circ}$
- (D) $10^2 = x^2 + y^2 - 2xy \cos 42^\circ$

10. For $y = \sqrt{1 - f(x)}$, $\frac{dy}{dx}$ is equal to

- (A) $\frac{2f'(x)}{\sqrt{1-f(x)}}$
- (B) $\frac{-1}{2\sqrt{1-f(x)}}$
- (C) $\frac{1}{2} \sqrt{1-f'(x)}$
- (D) $\frac{-f'(x)}{2\sqrt{1-f(x)}}$

Section 2

72 marks

Attempt Questions 11 – 18

Allow about 105 minutes for this section

Start each question on a new page

Question 11 (9 marks)

- a) Simplify $\frac{3p^2-3q^2}{6p-6q}$ 2
- b) If $\frac{\sqrt{2}}{9} = 2^m \times 3^n$ find the values of m and n . 2
- c) Solve $\frac{x+1}{x+3} = 5$ 2
- d) Differentiate $y = 4x^3 - 3x^2 - x + 2$ 2
- e) Find the exact value of $\tan 330^\circ$ 1

Question 12 (9 marks) Start a new page

- a) The points $A(1, 7)$, $B(-3, 5)$ and $C(4, -1)$ lie on a number plane.
- i) Find the length of AC. 1
- ii) Find the gradient of AC. 1
- iii) Find the equation of the line AC. 1
- iv) Find the perpendicular distance of B from the line AC. 1
- v) If ABCD is a parallelogram, find the coordinates of D. 1
- b) Find the exact solution of $2x^2 + 4x - 5 = 0$ 2
- c) Simplify $\sqrt{60} + (\sqrt{5} - \sqrt{3})^2$ 2

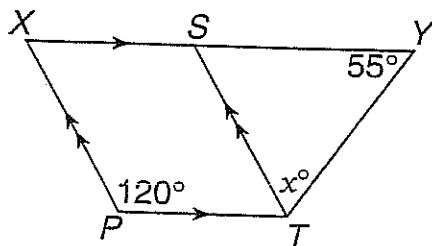
Question 13 (9 marks) Start a new page

a) Find the area bounded by the line $4x - y = 8$, the x axis and the y axis. 2

b) If $f(x) = x^2 + 2x$ find $\frac{f(x+h)-f(x)}{h}$ in simplest form. 2

c) Find the equation of the tangent to $y = (x - 3)^3$ at the point $(1, -8)$ 3

d) The diagram shows XY parallel to PT , XP parallel to ST ,
angle $XPT = 120^\circ$ and angle $SYT = 55^\circ$.



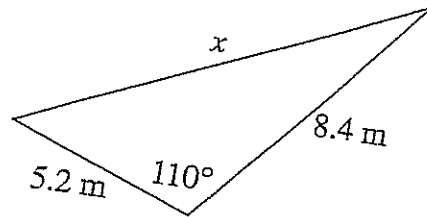
Find, with reasons, the value of x .

2

Question 14 (9 marks) Start a new page

- a) Find the value of x , correct to 1 decimal place.

2



- b) Solve $|2x - 4| < 2$

2

- c) Evaluate $\lim_{x \rightarrow 2} \frac{3x^2 - 5x - 2}{x - 2}$

2

- d) Indicate, by shading, the region where points simultaneously satisfy the inequalities $y \leq x^2 - 1$ and $x^2 + y^2 \leq 4$

3

Question 15 (9 marks) Start a new page

- a) If $\sin \theta = \frac{2}{3}$ and $\cos \theta < 0$ find the exact value of $\tan \theta$. 2
- b) Differentiate the following with respect to x
- i) $y = (5x - 3)^4$ 1
- ii) $y = \frac{6}{x^2}$ 1
- iii) $y = 12\sqrt{x^3}$ 1
- c) If the quadratic equation $ax^2 + bx + c = 0$ has a discriminant equal to 4, 2
what does this tell us about the nature of the roots of the equation ?
- d) If the lines $2x - 5y + 3 = 0$ and $ax + 4y + 12 = 0$ 2
are perpendicular, find the value of a .

Question 16 (9 marks) Start a new page

- a) Solve $\sin^2 \theta = \frac{3}{4}$ for $0 \leq \theta \leq 360^\circ$. 2
- b) If $f(x) = x\sqrt{2x+1}$ evaluate $f'(4)$. 2
- c) Find, correct to the nearest degree, the acute angle the line 2
 $3x - y - 3 = 0$ makes with the x axis.
- d) Evaluate $\lim_{x \rightarrow \infty} \frac{x^2-4}{2x^2+x}$ 1
- e) Simplify $\sin^3 A \sec A + \sin A \cos A$ 2

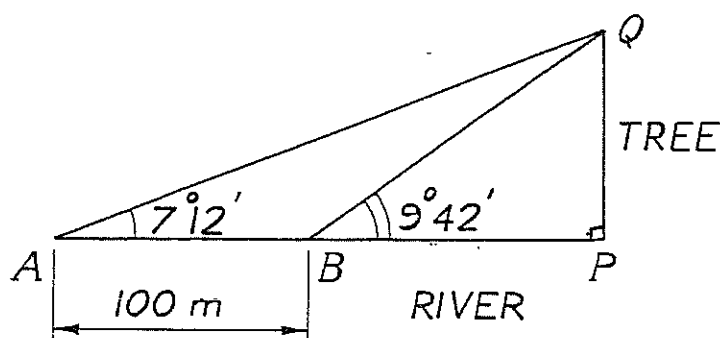
Question 17 (9 marks) Start a new page

a) State the domain of $y = \sqrt{x+4}$ 1

b) If $y = \frac{2x-1}{x+4}$ find $\frac{dy}{dx}$. 2

c) Simplify $\sin(90^\circ - \theta) \operatorname{cosec} \theta$ 2

d)



The diagram above was sketched by a surveyor, who measured the angle of elevation of a tree top on the other side of a river to be $7^\circ 12'$ at the point A. 4

At the point B, 100 metres directly towards the tree from A, the angle of elevation was $9^\circ 42'$.

Calculate the height of the tree, correct to 3 significant figures.

Please turn over

Question 18 (9 marks) Start a new page

- a) Find the gradient of the normal to the curve

$$y = x^2 + 6x + 3 \text{ at the point } (1, -2) \quad 2$$

- b) Find all values of k for which the quadratic equation 3

$$kx^2 - 8x + k = 0 \text{ has real roots.}$$

- c) The curve $y = ax + \frac{b}{x^2}$ cuts the x axis at the point $(2, 0)$ 4

and the gradient of the tangent to this curve at the point $(2, 0)$ equals 1.

Find the values of a and b .

End of paper

1. C

2. C

3. D

4. B

5. C

6. A

7. A

8. B

9. B

10. D

$$11. a. \frac{3(p-q)(p+q)}{6(p-q)}$$

$$= \frac{p+q}{2}$$

b. $m = \frac{1}{2}, n = -2$

c. $x+1 = 5x+15$

$4x = -14$

$x = \frac{-7}{2}$

d. $y' = 12x^2 - 6x - 1$

e. $-\frac{1}{\sqrt{3}}$

$$12. a. i. AC = \sqrt{(4-1)^2 + (-1-7)^2}$$

$$= \sqrt{73} \text{ units}$$

$$ii. m = \frac{7-1}{1-4}$$

$$= \frac{6}{-3}$$

$$iii. y-7 = -\frac{8}{3}(x-1)$$

$$3y-21 = -8x+8$$

$$8x+3y-29=0$$

$$iv. d = \frac{|-3 \times 8 + 5 \times 3 - 29|}{\sqrt{8^2 + 3^2}}$$

$$= \frac{38}{\sqrt{73}} \text{ units}$$

v. D (8,1)

$$b. x = \frac{-4 \pm \sqrt{4^2 - 4 \times 2 \times -5}}{4}$$

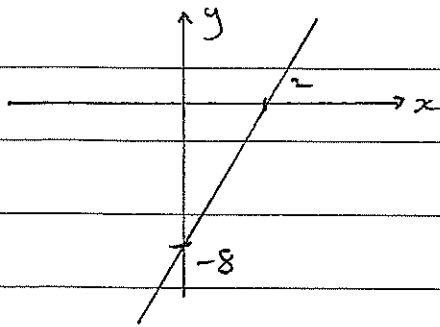
$$= \frac{-4 \pm \sqrt{56}}{4}$$

$$= \frac{-2 \pm \sqrt{14}}{2}$$

$$c. 2\sqrt{15} + 5 - 2\sqrt{15} + 3$$

$$= 8$$

13. a.



$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 8 \times 2 \\ &= 8 \text{ sq units} \end{aligned}$$

$$b. \frac{f(x+h) - f(x)}{h}$$

$$= \frac{(x+h)^2 + 2(x+h) - (x^2 + 2x)}{h}$$

$$= \frac{x^2 + 2xh + h^2 + 2x + 2h - x^2 - 2x}{h}$$

$$= 2x + h + 2$$

$$c. y' = 3(x-3)^2$$

$$\text{sub } x=1$$

$$\begin{aligned} m_T &= 3(1-3)^2 \\ &= 12 \end{aligned}$$

$$\therefore y + 8 = 12(x-1)$$

$$y + 8 = 12x - 12$$

$$y = 12x - 20$$

$$d. \angle XST = 120^\circ \text{ (opposite angles of a parallelogram)}$$

$$\therefore x = 65 \text{ (exterior angle of a triangle equals the sum of the opposite interior angles)}$$

$$14. a. x^2 = 5 \cdot 2^2 + 8 \cdot 4 - 2 \times 5 \cdot 2 \times 8 \cdot 4 \times \cos 110$$

$$x = 11.3 \text{ m}$$

$$b. -2 < 2x - 4 < 2$$

$$2 < 2x < 6$$

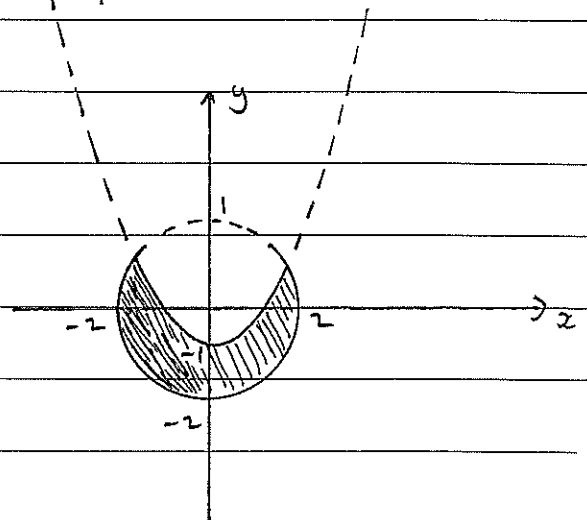
$$1 < x < 3$$

$$c. \lim_{x \rightarrow 2} \frac{(3x+1)(x-2)}{x-2}$$

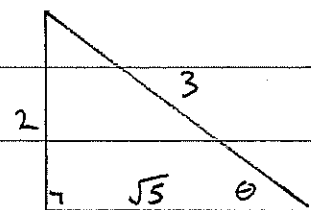
$$= \lim_{x \rightarrow 2} (3x+1)$$

$$= 7$$

d.



15. a.



2nd quadrant

$$\therefore \tan \theta = -\frac{2}{\sqrt{5}}$$

$$b. i. y' = 20(5x-3)^3$$

$$ii. y = 6x^{-2}$$

$$y' = -12x^{-3}$$

$$= \frac{-12}{x^3}$$

$$iii. y = 12x^{\frac{3}{2}}$$

$$y' = 12 \times \frac{3}{2} x^{\frac{1}{2}}$$

$$= 18\sqrt{x}$$

c. roots are real, different and rational

$$d. m_1 = \frac{2}{5} \quad m_2 = -\frac{9}{4}$$

$$\text{but } m_1 \times m_2 = -1$$

$$\frac{2}{5} \times -\frac{9}{4} = -1$$

$$2a = 20$$

$$a = 10$$

$$16. a. \sin \theta = \pm \frac{\sqrt{3}}{2}$$

$$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

$$b. f'(x) = \sqrt{2x+1} + \frac{x}{\sqrt{2x+1}}$$

$$f'(4) = \sqrt{9} + \frac{4}{\sqrt{9}}$$

$$= 4\frac{1}{3}$$

$$c. m = 3$$

$$\therefore \tan \theta = 3$$

$$\therefore \theta = 72^\circ$$

$$d. \frac{1}{2}$$

$$e. \sin A \left(\frac{\sin^2 A}{\cos A} + \cos A \right)$$

$$= \sin A \left(\frac{\sin^2 A + \cos^2 A}{\cos A} \right)$$

$$= \frac{\sin A}{\cos A}$$

$$= \tan A$$

$$17. a. x \geq -4$$

$$b. \frac{dy}{dx} = \frac{(x+4)(2) - (2x-1)(1)}{(x+4)^2}$$

$$= \frac{2x+8-2x+1}{(x+4)^2}$$

$$= \frac{9}{(x+4)^2}$$

$$c. \sin(90^\circ - \theta) \operatorname{cosec} \theta$$

$$= \frac{\cos \theta}{\sin \theta}$$

$$= \cot \theta$$

$$d. \angle AQB = 2^\circ 30'$$

$$\frac{BQ}{\sin 7^\circ 12'} = \frac{100}{\sin 2^\circ 30'}$$

$$BQ = \frac{100 \sin 7^\circ 12'}{\sin 2^\circ 30'}$$

$$\text{and } \sin 9^\circ 42' = \frac{h}{BQ}$$

$$\therefore h = \frac{100 \sin 7^\circ 12' \sin 9^\circ 42'}{\sin 2^\circ 30'}$$

$$= 48.4 \text{ m}$$

$$18. a. y' = 2x + 6$$

$$\text{when } x = 1$$

$$m_T = 8$$

$$\therefore m_N = -\frac{1}{8}$$

$$b. \text{ real roots } \Rightarrow \Delta \geq 0$$

$$\therefore (-8)^2 - 4 \times k \times k \geq 0$$

$$64 - 4k^2 \geq 0$$

$$k^2 \leq 16$$

$$-4 \leq k \leq 4$$

$$c. (2, 0) \text{ satisfies } y = ax + \frac{b}{x^2}$$

$$\therefore 0 = 2a + \frac{b}{4}$$

$$\text{or } 8a + b = 0$$

$$y = ax + bx^{-2}$$

$$y' = a - 2bx^{-3}$$

$$\text{when } x = 2 \quad y' = 1$$

$$\therefore 1 = a - \frac{2b}{8}$$

$$8 = 8a - 2b$$

$$4 = 4a - b$$

$$\therefore \text{Solve simultaneously}$$

$$8a + b = 0$$

$$4a - b = 4 \quad \text{add}$$

$$12a = 4$$

$$a = \frac{1}{3}$$

$$\therefore b = -\frac{8}{3}$$